

NOTICE & COMPLIANCE STATEMENT

This dataset is completely synthetic and fictional. It was created solely for demonstrating the IBM i Incident Insight & Response Assistant Capstone prototype. It does not contain client, production, personal, or confidential information.

TECHNICAL SPECIFICATION & DATA DICTIONARY

IBM i Incident Insight & Response Assistant

Operational Incident Schema Specification & Engineering Variable Reference

TARGET DATASET	TEMPORAL WINDOW	TOTAL OBSERVATIONS	PRIMARY APPLICATION
IBM_i_Incident_Data.xlsx	1 Calendar Month (30 Days)	580 Logged Incidents	Deterministic KPIs & AI Insights

This data dictionary formally specifies the 13 feature columns comprising the incident telemetry dataset. Each record represents an operational alert or subsystem event captured from the IBM i system operator message queues (QSYSOPR), licensed internal code logs, and performance management monitors.

COLUMN NAME	DATA TYPE	ALLOWED / SAMPLE VALUES	DESCRIPTION & OPERATIONAL MEANING	CONSTRAINTS & RULES
incident_id PRIMARY KEY	String (Alphanumeric)	INC0001, INC0002, ..., INC0580	Unique identifier generated sequentially for every logged IBM i operational incident ticket.	Unique, non-null, immutable. Regex: ^INC\d{4}\$.
timestamp DATETIME	Datetime (ISO 8601)	YYYY-MM-DD HH:MM:SS (e.g. 2024-06-08 03:14:22)	Timestamp when the diagnostic threshold violation or OS exception was recorded in the system message queue.	Spans 30 complete days. Captures overnight batch windows & peak business hours.
system_name CATEGORICAL	String (Categorical)	IBMI-PROD-01, IBMI-PROD-02, IBMI-DR-01, IBMI-TEST-01	Logical partition (LPAR) or host partition where the operational alert or failure condition originated.	Production partitions host core transaction workloads; TEST-01 has lowest incident count.
subsystem CATEGORICAL	String (Categorical)	QBATCH, QINTER, QSERVER, QSYSWRK, QSPL	IBM i operating execution environment controlling resource allocations and routing entries for the job.	Maps directly to standard OS/400 workload management architectures.
alert_category CATEGORICAL	String (Categorical)	PERFORMANCE, WORK_MGMT, STORAGE, SECURITY, SPOOL, NETWORK	Functional operational domain classifying the telemetry condition or system alert.	Used for deterministic KPI aggregation and pattern clustering.
message_id STRING	String (7-Char Code)	CPI0999, CPF1124, CPI099B, CPF2234, CPF3303, CPI9162, etc.	Standard IBM i OS message identifier representing the specific licensed internal code or exception identifier.	3-character subsystem prefix (CPI/CPF/TCP) + 4 alphanumeric diagnostic characters.
severity CATEGORICAL	String (Ordinal)	Low, Medium, High, Critical	Technical severity rating evaluating the degree of hardware, OS, or queue resource degradation.	Assessed by automated system monitor independently of immediate commercial impact.

COLUMN NAME	DATA TYPE	ALLOWED / SAMPLE VALUES	DESCRIPTION & OPERATIONAL MEANING	CONSTRAINTS & RULES
description TEXT	String (Free Text)	Concise technical failure summary	Technical description of the trigger threshold, affected job queue, or operating system status.	Contains operational symptoms only; explicitly excludes pre-diagnosed root causes.
resolution_status CATEGORICAL	String (Categorical)	Resolved, In Progress, Investigating, New	Current engineering lifecycle status of the incident ticket in the operational tracking queue.	Majority are Resolved (~90%); active backlog incidents remain open without resolution duration.
resolution_duration_minutes METRIC	Integer (Minutes)	0 (unresolved), 8 to 1480 (resolved)	Total elapsed time in minutes from incident generation to technical verification and closure.	Strictly 0 for unresolved statuses (New, Investigating, In Progress); >0 for Resolved.
business_impact CATEGORICAL	String (Ordinal)	Low, Medium, High, Critical	Organizational and commercial disruption scale measuring productivity loss and SLA exposure.	Correlates positively with severity, but accounts for partition role (e.g., TEST-01 impact is capped).
affected_users METRIC	Integer (Count)	0 to 1420 users	Estimated volume of interactive users, client interfaces, or active batch job profiles disrupted.	Positively correlated with business impact; includes rare high-volume enterprise lock outliers.
assigned_team CATEGORICAL	String (Categorical)	IBM i Operations, Performance Support, Storage Support, Security Support, Network Support	Specialized Tier-2/Tier-3 engineering squad assigned ownership of triage and incident resolution.	Logically mapped to alert category domains with realistic Tier-1 triage exceptions.

ARCHITECTURAL SUBSYSTEM REFERENCE GUIDE

- **QBATCH:** Dedicated batch execution subsystem handling background processing, data warehouse ingestion, and end-of-day billing.
- **QINTER:** Interactive subsystem supporting 5250 terminal emulation sessions, user workstations, and foreground menu navigation.
- **QSERVER:** Host server daemons managing database access (ODBC/JDBC), client APIs, remote commands, and distributed transactions.
- **QSYSWRK:** System work subsystem hosting system daemons, hardware monitors, event listeners, and background OS management tasks.
- **QSPL:** Spooling subsystem coordinating print writers, output queues, and job log spooled file life cycles.